DATA STORAGE SOLUTIONS FOR DATA ANALYTICS

Data Warehouse Design, Implementation and BI Reporting

Module Code: B9DA111

Group Members:

Motupalli Kota Naga Satya Hemanth - 20069760

Poongundran Elanthiraiyan - 20087673

Sana JN Manikanta Swamy - 200773371

Siva Kumar Thandra - 20026949

Data Source: U.S. Department of Transportation (BTS Airline On-Time Performance Data)

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1. INTRODUCTION

This project presents a proof-of-concept data warehouse designed to analyse airline arrival performance data. The system integrates operational airline delay data into a structured dimensional model to support business intelligence reporting and visualization.

The source dataset is obtained from the U.S. Department of Transportation (Bureau of Transportation Statistics) Airline On-Time Performance dataset, which includes monthly airline arrival statistics such as delays, cancellations, diversions, and delay causes.

The goal of this data warehouse is to enable analytical reporting on airline performance trends, airport efficiency, and carrier performance.

Source:

U.S. Department of Transportation BTS Airline On-Time Performance Dataset
https://www.transtats.bts.gov/OT_Delay/OT_DelayCause1.asp?20=E

1.2 REASONS FOR SELECTING THE SUBJECT AREA AND DATA

Airline performance analysis is a highly relevant business domain due to its impact on customer satisfaction, operational efficiency, and cost management.

This dataset was selected because:

- It contains structured time-series operational data
- It includes multiple analytical dimensions (carrier, airport, time)
- It provides measurable KPIs (delays, cancellations, diversions)
- It supports BI reporting and trend analysis

The subject area is suitable for building a star schema data warehouse.

1.3 VISION AND GOALS

The vision of this project is to design a scalable data warehouse that enables airline performance monitoring and decision-making.

Goals:

- Analyse airport performance based on delays
- Identify worst-performing airports
- Compare airline carriers
- Analyse monthly trends
- Support BI dashboards and reporting

1.4 KEY STAKEHOLDERS

- Airline Operations Managers
- Airport Authorities
- Aviation Analysts
- Government Transport Agencies

- Data Analysts and BI Developers

1.5 BUSINESS REQUIREMENTS

The system must support the following requirements:

- Total flights per airport and carrier
- Delay analysis by cause (weather, carrier, NAS, security)
- Monthly trend analysis of delays
- Identification of top 10 worst airports
- Carrier performance comparison
- KPI: Delay rate per carrier. Delay Rate KPI formula: Delay Rate = Flights Delayed / Total Flights

1.6 ADDITIONAL SCOPE

The project also includes:

- SQL Server Data Warehouse implementation
- SSRS reporting
- Tableau dashboard
- Neo4j graph database comparison

2. SCHEMA DESIGN

The data warehouse follows a Star Schema design.

Fact Table:

- Arrival_Fact

Dimension Tables:

- Dim_Date
- Dim_Airport
- Dim_Carrier

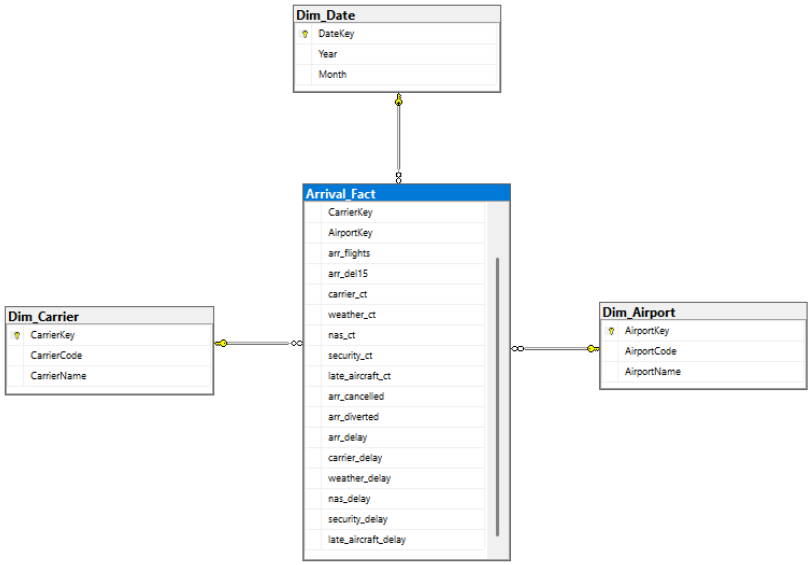
Staging Table:

- Airline_Staging

Design Justification:

- Star schema improves query performance
- Simplifies BI reporting
- Reduces joins complexity
- Suitable for OLAP workloads

Schema Diagram



3. ETL PROCESS

The ETL process involves:

Extract:

- Data extracted from BTS airline dataset (CSV)

Staging Layer:

- Raw data is first loaded into staging tables
- Used to store unprocessed data before transformation
- Helps in data validation and error handling

Transform:

- Cleaning null values
- Formatting numeric fields
- Mapping carrier and airport codes
- Creating surrogate keys

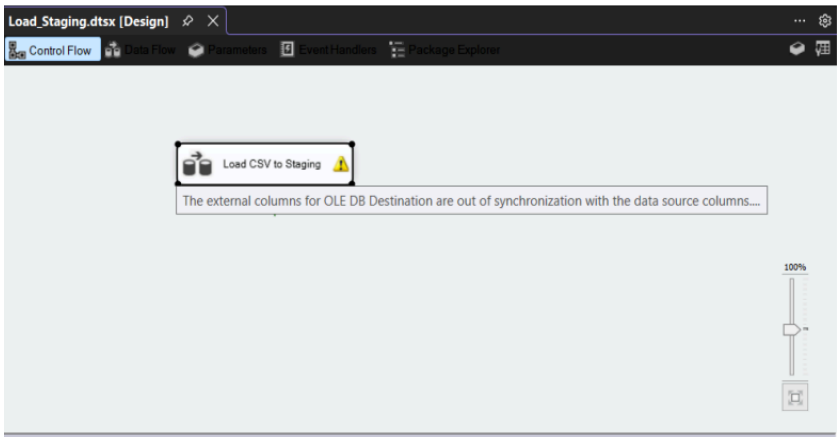
Load:

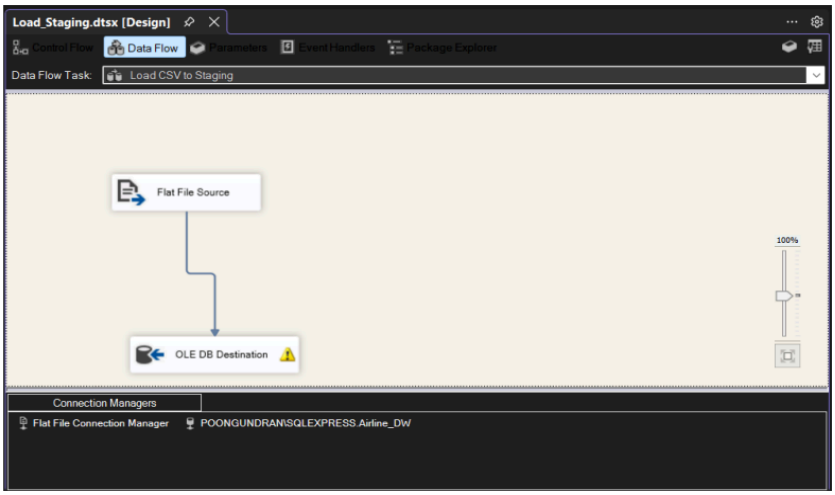
Data loaded into:

- Dimension tables first
- Fact table after establishing foreign key relationships

Tools Used:

- SQL Server Management Studio (SSMS)
- SQL Server Integration Services (SSIS)





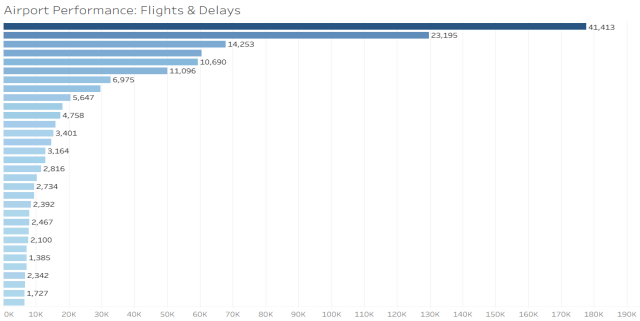
4. VISUALIZATIONS AND REPORTS

4.1 VISUALIZATIONS (TABLEAU)

A Tableau dashboard was created with the following visuals:

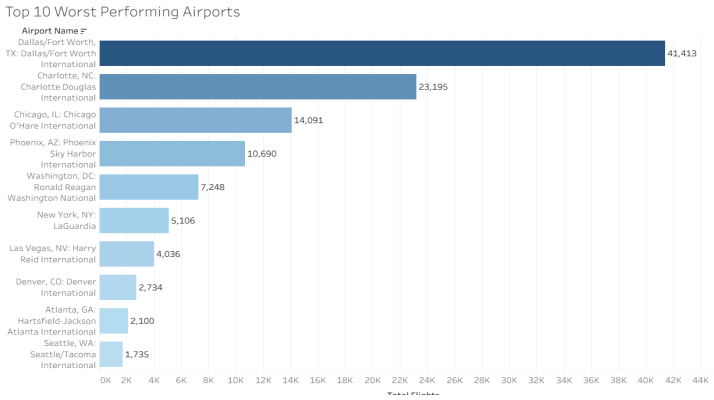
1. Airport Performance (Flights & Delays)

- Horizontal bar chart
- Shows total flights vs delays per airport



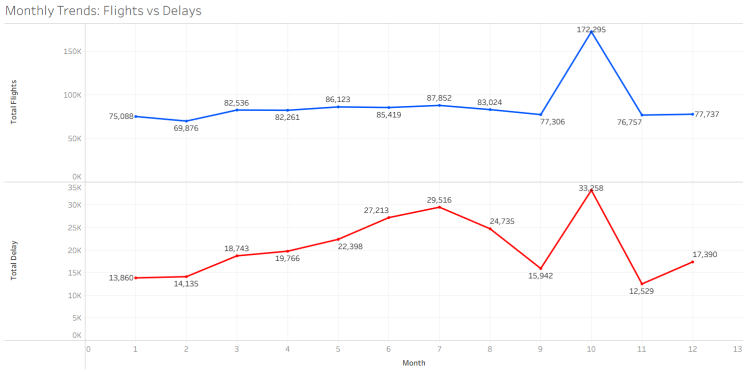
2. Top 10 Worst Performing Airports

- Horizontal bar chart
- Based on total delays



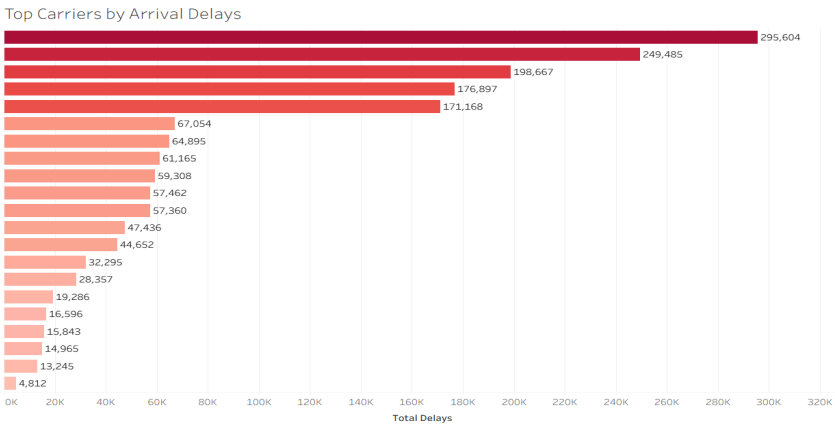
3. Monthly Trends (Flights & Delays)

- Line chart
- Shows trend over time



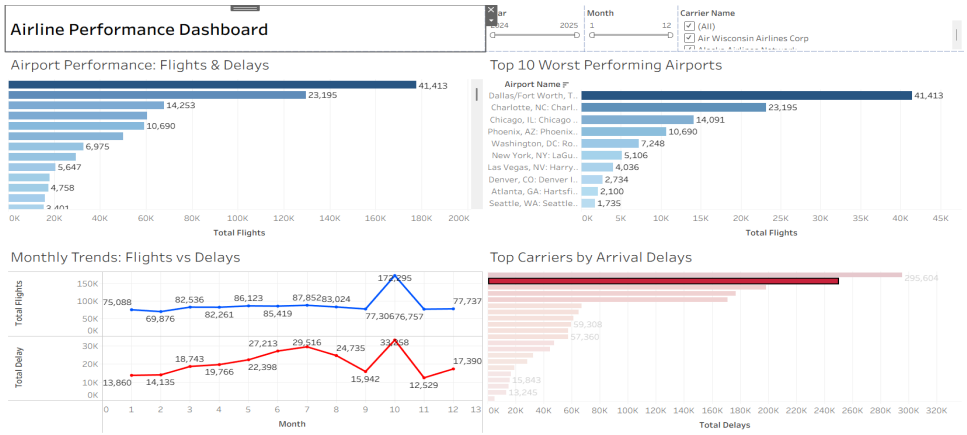
4. Top Carriers by Delays

- Horizontal bar chart
- Carrier comparison



Dashboard

A Tableau dashboard combines all visualizations.



4.2 SSRS REPORTS

Four SSRS reports were developed:

Report 1: Flight Summary by Carrier

- Total flights
- Delays
- Cancellations
- Delay percentage

Flight Volume and Delay Summary by Carrier

Carrier Name	Year	Month	Total Flights	Flights Delayed	Cancelled Flights	Diverted Flights	Total Delay Minutes	Pct Flights Delayed
Air Wisconsin Airlines Corp	2025	4	264	123	31	2	16,295	46.59%
CommuteAir LLC dba CommuteAir	2024	10	6,401	534	8	9	39,852	8.34%
Mesa Airlines Inc.	2025	9	6,383	564	0	11	39,535	8.84%
Air Wisconsin Airlines Corp	2024	10	4,301	573	0	3	46,926	13.32%
Air Wisconsin Airlines Corp	2024	11	3,642	581	22	1	50,964	15.95%
Mesa Airlines Inc.	2024	10	7,238	651	1	4	44,502	8.99%
Hawaiian Airlines Network	2024	11	6,427	662	29	5	27,691	10.30%
CommuteAir LLC dba CommuteAir	2024	11	6,328	677	4	6	52,199	10.70%
CommuteAir LLC dba CommuteAir	2025	9	6,413	691	0	15	50,168	10.77%
Mesa Airlines Inc.	2024	11	6,763	701	2	11	43,515	10.37%
Hawaiian	2025	9	6,428	743	11	5	39,969	11.56%

Report 2: Airport Performance Report

- Flights and delay causes by airport

Airport Performance

Airport	Airport Name	Total Flights	Total Delays	Carrier Delays	Weather Delays	NASDelays	Late Aircraft Delays
ORD	Chicago, IL: Chicago O'Hare International	377,495	84,752	24,425	3,398	31,159	25,688
DFW	Dallas/Fort Worth, TX: Dallas/Fort Worth International	344,993	78,407	21,108	5,677	22,534	28,928
DEN	Denver, CO: Denver International	344,416	74,734	21,618	3,040	29,195	20,776
ATL	Atlanta, GA: Hartsfield-Jackson Atlanta Intern	378,642	66,110	19,301	2,417	17,704	26,618
CLT	Charlotte, NC: Charlotte Douglas International	267,713	49,262	12,626	2,456	12,422	21,655
LAS	Las Vegas, NV: Harry Reid Internati	205,850	40,569	10,873	1,120	12,440	16,049
DCA	Washington, DC: Ronald Reagan	159,380	39,231	8,377	1,265	15,890	13,635

Report 3: Top 10 Worst Airports

- Ranked by total delays

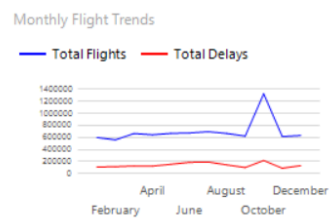
Airport	Airport Name	Total Flights	Total Delays	Carrier Delays	Weather Delays	NASDelays	Late Aircraft Delays
ORD	Chicago, IL: Chicago O'Hare International	377495	84752	24425	3398.22	31159.37	25688.11
DFW	Dallas/Fort Worth, TX: Dallas/Fort Worth International	344993	78407	21107.79	5677.08	22533.73	28927.53
DEN	Denver, CO: Denver International	344416	74734	21618.32	3039.98	29195.08	20775.67
ATL	Atlanta, GA: Hartsfield-Jackson Atlanta Intern	378642	66110	19300.51	2416.57	17703.89	26617.79
CLT	Charlotte, NC: Charlotte Douglas International	267713	49262	12626.29	2455.94	12421.52	21654.72
LAS	Las Vegas, NV: Harry Reid Internati	205850	40569	10872.61	1120.19	12440.47	16048.78
DCA	Washington, DC: Ronald Reagan Washington National	159380	39231	8376.63	1265.46	15889.92	13635.32
LGA	New York, NY:	179308	38546	9658.08	1421.73	16054.36	11352.27

Report 4: Monthly Trends Report

- Time-based trend analysis

Monthly Arrival Delays Trends

Year	Month	Total Flights	Total Delays	Carrier Delays	Weather Delays	NAS Delays	Late Aircraft Delays
2024	10	656,283	84,806	31,555	1,472	22,390	29,180
2024	11	614,597	89,260	30,152	2,569	27,289	29,060
2024	12	631,944	132,515	43,489	5,392	34,704	48,621
2025	1	599,013	109,605	35,995	5,284	32,859	35,193
2025	2	559,577	114,715	34,221	4,823	36,428	39,050
2025	3	664,932	127,648	39,034	4,596	37,272	46,529
2025	4	644,084	124,749	37,564	4,663	36,994	45,297
2025	5	667,586	155,262	43,212	7,216	49,775	54,855
2025	6	674,179	186,766	55,228	8,968	50,277	71,962
2025	7	696,049	192,888	57,601	9,663	50,585	74,731
2025	8	666,242	145,517	45,479	6,894	38,191	54,684
2025	9	621,601	100,369	32,171	3,670	30,020	34,303
2025	10	668,332	132,452	40,208	2,699	41,161	48,172



5. GRAPH DATABASE (NEO4J)

A graph database was implemented using Neo4j to model relationships between:

- Flights
- Airports
- Carriers

Example Structure:

- (Carrier)-[:OPERATES]->(Flight)
- (Flight)-[:ARRIVES_AT]->(Airport)

Neo4j Cypher Queries

1. Total Flights per Airport

```
MATCH (f:Flight)-[:ARRIVES_AT]->(a:Airport)
RETURN a.Name AS Airport, SUM(f.ArrFlights) AS TotalFlights
ORDER BY TotalFlights DESC;
```

2. Total Delays per Airport

```
MATCH (f:Flight)-[:ARRIVES_AT]->(a:Airport)
RETURN a.Name AS Airport, SUM(f.ArrDel15) AS TotalDelays
ORDER BY TotalDelays DESC;
```

3. Total Flights per Carrier

```
MATCH (c:Carrier)-[:OPERATES]->(f:Flight)
RETURN c.Name AS Carrier, SUM(f.ArrFlights) AS TotalFlights
ORDER BY TotalFlights DESC;
```

4. Total Delays per Carrier

```
MATCH (c:Carrier)-[:OPERATES]->(f:Flight)
RETURN c.Name AS Carrier, SUM(f.ArrDel15) AS TotalDelays
ORDER BY TotalDelays DESC;
```

5. Monthly Flight Volume

```
MATCH (f:Flight)
RETURN f.Date AS Month, SUM(f.ArrFlights) AS TotalFlights
ORDER BY Month;
```

6. Monthly Delay Trends

```
MATCH (f:Flight)
RETURN f.Date AS Month, SUM(f.ArrDel15) AS TotalDelays
ORDER BY Month;
```

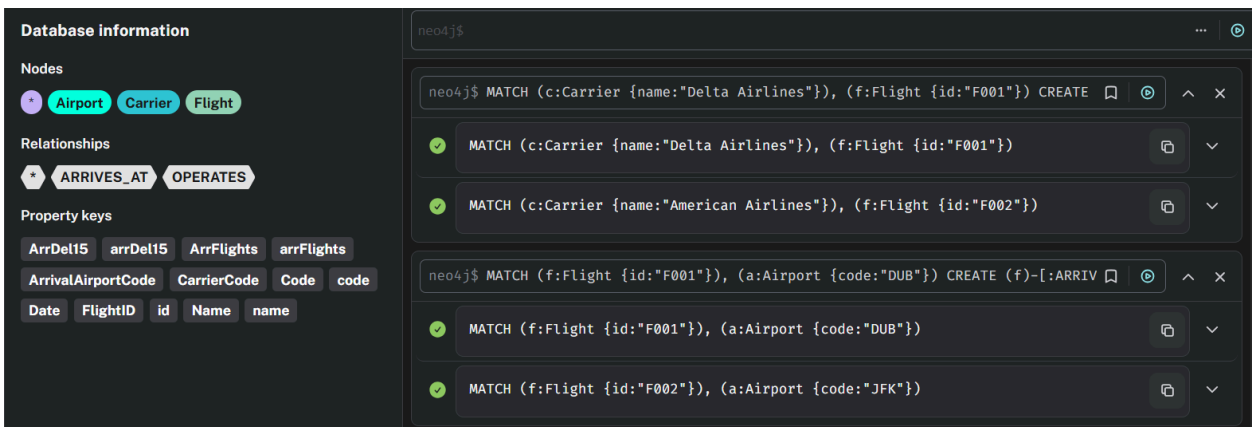
7. Delay Rate per Carrier

```
MATCH (c:Carrier)-[:OPERATES]->(f:Flight)
RETURN c.Name AS Carrier,
    SUM(f.ArrDel15) * 1.0 / SUM(f.ArrFlights) AS DelayRate
ORDER BY DelayRate DESC;
```

A simplified subset of airline data was modelled in Neo4j using sample airports and carriers to demonstrate graph relationships between entities. This approach was used to improve clarity of visualization and support proof-of-concept analysis rather than loading the full BTS dataset.

Nodes

Graph with connections



Airport performance results

```
MATCH (f:Flight)-[:ARRIVES_AT]->(a:Airport)
RETURN a.name AS Airport,
       SUM(f.arrFlights) AS TotalFlights,
       SUM(f.arrDel15) AS TotalDelays
ORDER BY TotalDelays DESC;
```

Table RAW

	Airport	TotalFlights	TotalDelays
1	"John F. Kennedy Airport"	200	50
2	"Dublin Airport"	120	30

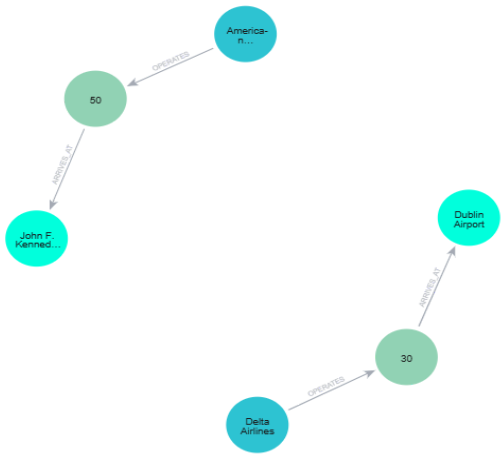
Carrier performance results

```
neo4j$ MATCH (c:Carrier)-[:OPERATES]->(f:Flight) RETURN c.name AS Carrier, S
```

Table RAW

	Carrier	TotalFlights	TotalDelays
1	"American Airlines"	200	50
2	"Delta Airlines"	120	30

Relationship graph



5.1 COMPARISON TO RELATIONAL DATABASES

Relational Database (SQL Server):

- Structured tables
- Uses joins for relationships
- Good for transactional processing
- Slower for deep relationship queries

Graph Database (Neo4j):

- Relationship-based model
- Faster traversal of connections

- Ideal for network-based queries
- More intuitive for connected data

Key Difference:

Neo4j performs faster for relationship-heavy queries like:

- airport → flights → carriers

while SQL requires multiple joins.

6. CONCLUSION

This project successfully demonstrates the design and implementation of a data warehouse for airline performance analysis.

The system provides:

- Efficient star schema design
- ETL pipeline implementation
- BI dashboards and SSRS reports
- Graph database comparison

The combination of SQL Server, Tableau, SSRS, and Neo4j provides a complete business intelligence solution.

7. BIBLIOGRAPHY

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  https://www.transtats.bts.gov/
- Kimball, R. (2013) The Data Warehouse Toolkit
- Microsoft SQL Server Documentation
  https://learn.microsoft.com/sql/
- Neo4j Graph Database Documentation
  https://neo4j.com/docs/
- Tableau Official Documentation
  https://help.tableau.com/

APPENDIX A - VISUALIZATION CODE (SQL QUERIES / SSRS)

The following stored procedures were created in SQL Server Management Studio and used as data sources for SSRS reporting. Each procedure was executed to generate business intelligence outputs.

1. Flight Summary by Carrier

```
USE [Airline_DW]
GO
/****** Object:  StoredProcedure [dbo].[sp_FlightSummaryByCarrier_Report1]    Script Date:
11-04-2026 09:29:55 ******/
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
ALTER PROCEDURE [dbo].[sp_FlightSummaryByCarrier_Report1]
AS
BEGIN
   SET NOCOUNT ON;

   SELECT
      c.CarrierName,
      d.Year,
      d.Month,
      SUM(f.arr_flights) AS TotalFlights,
      SUM(f.arr_del15) AS FlightsDelayed,
      SUM(f.arr_cancelled) AS CancelledFlights,
      SUM(f.arr_diverted) AS DivertedFlights,
      SUM(f.arr_delay) AS TotalDelayMinutes,

      CAST(SUM(f.arr_del15) AS FLOAT) /
      NULLIF(SUM(f.arr_flights), 0) AS PctFlightsDelayed

   FROM dbo.Arrival_Fact f
   INNER JOIN dbo.Dim_Carrier c
      ON f.CarrierKey = c.CarrierKey
   INNER JOIN dbo.Dim_Date d
      ON f.DateKey = d.DateKey
   GROUP BY c.CarrierName, d.Year, d.Month
```

```
    ORDER BY d.Year, d.Month, c.CarrierName;
END
```

SQLQuery1.s...poong (53)) | poongundra..._DW_Diagram

```
USE [Airline_DW]
GO
/****** Object:  StoredProcedure [dbo].[sp_FlightSummaryByCarrier_Report1]    Script Date: 11-04-2026 09:29:55 ******/
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
ALTER PROCEDURE [dbo].[sp_FlightSummaryByCarrier_Report1]
AS
BEGIN
    SET NOCOUNT ON;

    SELECT
        c.CarrierName,
        d.Year,
        d.Month,
        SUM(f.arr_flights) AS TotalFlights,
        SUM(f.arr_del15) AS FlightsDelayed,
        SUM(f.arr_cancelled) AS CancelledFlights,
        SUM(f.arr_diverted) AS DivertedFlights,
        SUM(f.arr_delay) AS TotalDelayMinutes,

        CAST(SUM(f.arr_del15) AS FLOAT) /
        NULLIF(SUM(f.arr_flights), 0) AS PctFlightsDelayed

    FROM dbo.Arrival_Fact f
    INNER JOIN dbo.Dim_Carrier c
        ON f.CarrierKey = c.CarrierKey
    INNER JOIN dbo.Dim_Date d
        ON f.DateKey = d.DateKey
    GROUP BY c.CarrierName, d.Year, d.Month
    ORDER BY d.Year, d.Month, c.CarrierName;
END
```

SQLQuery2....poong (82))* | SQLQuery1.s...\poong (53)) | poongundra..._DW_Diagram

```
EXEC sp_FlightSummaryByCarrier_Report1;
```

110 % | No issues found | Ln: 1, Ch: 26 | TABS | CRLF | UTF-

Results | Messages

	CarrierName	Year	Month	TotalFlights	FlightsDelayed	CancelledFlights	DivertedFlights	TotalDelayMinutes	PctFlightsDelayed
1	Air Wisconsin Airlines Corp	2024	10	4301	573	0	3	46926	0.13322483143455
2	Alaska Airlines Network	2024	10	20503	3436	174	42	156761	0.167585231429547
3	Allegiant Air	2024	10	9195	1304	588	19	112689	0.141816204458945
4	American Airlines Network	2024	10	86812	13578	766	91	952209	0.156406948348155
5	CommuteAir LLC dba CommuteAir	2024	10	6401	534	8	9	39852	0.0834244649273551
6	Delta Air Lines Network	2024	10	88300	8428	799	85	506286	0.0954473386183465
7	Endeavor Air Inc.	2024	10	16843	1326	0	8	85574	0.0787270676245325
8	Envoy Air	2024	10	23564	2506	64	12	130356	0.106348667458836
9	Frontier Airlines	2024	10	17801	4185	555	5	301398	0.235099151733049
10	GoJet Airlines LLC d/b/a United Express	2024	10	5394	749	0	7	51546	0.138857990359659
11	Hawaiian Airlines Network	2024	10	6583	839	73	7	34465	0.127449491113474
12	Horizon Air	2024	10	7248	945	29	8	45035	0.130380794701987
13	JetBlue Airways	2024	10	19694	2717	489	33	158303	0.137960800243729
14	Mesa Airlines Inc.	2024	10	7238	651	1	4	44502	0.0899419729206963
15	Piedmont Airlines	2024	10	10199	1002	6	17	59817	0.0982449259731346
16	PSA Airlines Inc.	2024	10	21144	3209	102	26	234833	0.151768823306848
17	Republic Airline	2024	10	27176	2208	34	22	114708	0.0812481601413012
18	SkyWest Airlines Inc.	2024	10	69057	9887	108	115	732904	0.143171582895289
19	Southwest Airlines	2024	10	119251	15707	1546	135	720821	0.131713780177944
20	Spirit Airlines	2024	10	21176	3070	421	22	196588	0.144975443898753
21	United Air Lines Network	2024	10	68403	7952	551	77	462259	0.116252211160329
22	Air Wisconsin Airlines Corp	2024	11	3642	581	22	1	50964	0.159527732015376
23	Alaska Airlines Network	2024	11	19462	3270	115	36	150230	0.168019730757373
24	Allegiant Air	2024	11	8757	1379	5	10	101842	0.157474020783373
25	American Airlines Network	2024	11	76757	12529	491	157	896458	0.163229412301158
26	CommuteAir LLC dba CommuteAir	2024	11	6328	677	4	6	52199	0.106984829329962
27	Delta Air Lines Network	2024	11	81825	9182	24	95	538994	0.112215093186679
28	Endeavor Air Inc.	2024	11	16021	1551	4	18	95965	0.0968104363023532
29	Envoy Air	2024	11	21821	2824	185	40	170531	0.129416617020302
30	Frontier Airlines	2024	11	17559	3914	151	19	284909	0.222905632439205
31	GoJet Airlines LLC d/b/a United Express	2024	11	5558	1435	40	14	137827	0.258186397984887
32	Hawaiian Airlines Network	2024	11	6427	662	29	5	27691	0.103002956278201
33	Horizon Air	2024	11	7008	922	105	19	45740	0.131563926940639
34	JetBlue Airways	2024	11	19134	3001	0	25	181816	0.156841225044424
35	Mesa Airlines Inc.	2024	11	6763	701	2	11	43515	0.103652225343782

2. Airport Performance Report

```
USE [Airline_DW]
GO
/****** Object:  StoredProcedure [dbo].[sp_AirportPerformance_Report2]    Script Date: 11-04-2026 09:36:42 ******/
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
ALTER PROCEDURE [dbo].[sp_AirportPerformance_Report2]
AS
BEGIN
    SELECT
```

```
    a.AirportCode AS Airport,
    a.AirportName,
    SUM(f.arr_flights) AS TotalFlights,
    SUM(f.arr_del15) AS TotalDelays,
    SUM(f.carrier_ct) AS CarrierDelays,
    SUM(f.weather_ct) AS WeatherDelays,
    SUM(f.nas_ct) AS NASDelays,
    SUM(f.late_aircraft_ct) AS LateAircraftDelays
  FROM dbo.Arrival_Fact f
  JOIN dbo.Dim_Airport a
    ON f.AirportKey = a.AirportKey
  GROUP BY
    a.AirportCode,
    a.AirportName
  ORDER BY
    TotalDelays DESC;
END
```

SQLQuery3.s...poong (85)) | SQLQuery2.s...\poong (82))* | SQLQuery1.s...\poong (53)) | poongundra..._DW_Diagram

```
USE [Airline_DW]
GO
/****** Object:  StoredProcedure [dbo].[sp_AirportPerformance_Report2]    Script Date: 11-04-2026 09:36:42 ******/
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
ALTER PROCEDURE [dbo].[sp_AirportPerformance_Report2]
AS
BEGIN
    SELECT
        a.AirportCode AS Airport,
        a.AirportName,
        SUM(f.arr_flights) AS TotalFlights,
        SUM(f.arr_del15) AS TotalDelays,
        SUM(f.carrier_ct) AS CarrierDelays,
        SUM(f.weather_ct) AS WeatherDelays,
        SUM(f.nas_ct) AS NASDelays,
        SUM(f.late_aircraft_ct) AS LateAircraftDelays
    FROM dbo.Arrival_Fact f
    JOIN dbo.Dim_Airport a
        ON f.AirportKey = a.AirportKey
    GROUP BY
        a.AirportCode,
        a.AirportName
    ORDER BY
        TotalDelays DESC;
END
```

```
EXEC sp_AirportPerformance_Report2;
```

110 % | No issues found | Ln: 1, C

Results | Messages

	Airport	AirportName	TotalFlights	TotalDelays	CarrierDelays	WeatherDelays	NASDelays	LateAircraftDelays
1	ORD	Chicago, IL: Chicago O'Hare International	377495	84752	24425	3398.22	31159.37	25688.11
2	DFW	Dallas/Fort Worth, TX: Dallas/Fort Worth International	344993	78407	21107.79	5677.08	22533.73	28927.53
3	DEN	Denver, CO: Denver International	344416	74734	21618.32	3039.98	29195.08	20775.67
4	ATL	Atlanta, GA: Hartsfield-Jackson Atlanta International	378642	66110	19300.51	2416.57	17703.89	26617.79
5	CLT	Charlotte, NC: Charlotte Douglas International	267713	49262	12626.29	2455.94	12421.52	21654.72
6	LAS	Las Vegas, NV: Harry Reid International	205850	40569	10872.61	1120.19	12440.47	16048.78
7	DCA	Washington, DC: Ronald Reagan Washington National	159380	39231	8376.63	1265.46	15889.92	13635.32
8	LGA	New York, NY: LaGuardia	179308	38546	9658.08	1421.73	16054.36	11352.27
9	SEA	Seattle, WA: Seattle/Tacoma International	199957	38379	10677.16	1426.86	12515.09	13557.46
10	PHX	Phoenix, AZ: Phoenix Sky Harbor International	214106	38203	13123.39	1144.48	7977.31	15880.75
11	BOS	Boston, MA: Logan International	158198	37893	9073.68999999999	920.27	16522.96	11329.24
12	EWR	Newark, NJ: Newark Liberty International	145323	37713	6901.04	1038.4	21695.24	8057.91
13	MCO	Orlando, FL: Orlando International	171124	37526	10518.79	889.71	13127.76	12917.65
14	LAX	Los Angeles, CA: Los Angeles International	213985	37244	14159.95	1511.66	8866.13	12617.69
15	SFO	San Francisco, CA: San Francisco International	160803	37223	10952.45	1105.08	15105.24	10005.62
16	IAH	Houston, TX: George Bush Intercontinental/Houston	183831	33593	10179.5	1257.01	12429.6	9703.39
17	PHL	Philadelphia, PA: Philadelphia International	135943	30356	7864.92	1099.23	9095.59000000001	12253.52
18	JFK	New York, NY: John F. Kennedy International	135421	27466	7690.88	799.37	10842.92	8082.56
19	DTW	Detroit, MI: Detroit Metro Wayne County	153198	26682	10190.84	1816.13	5476.52999999999	9162.17
20	MIA	Miami, FL: Miami International	119789	25736	7589.12	890.61	8781.79	8396.45
21	SAN	San Diego, CA: San Diego International	106392	25093	7441.61	732.38	8099.85	8768.23
22	MSP	Minneapolis, MN: Minneapolis-St Paul International	140328	23620	8487.85	1480.78	5510.11	8105.87
23	BNA	Nashville, TN: Nashville International	115522	23115	6839.36	719.89	5815.39	9684.93
24	FLL	Fort Lauderdale, FL: Fort Lauderdale-Hollywood Inter...	93414	20391	6054.51	515.29	7169.28	6585.5
25	AUS	Austin, TX: Austin - Bergstrom International	94037	19636	6278.84	712.58	5378.11	7238.85
26	BWI	Baltimore, MD: Baltimore/Washington International T...	104840	19152	5637.49	503.35	4812.26	8131.32
27	TPA	Tampa, FL: Tampa International	85382	19079	5794.67	453.91	5501.56	7294.48
28	SLC	Salt Lake City, UT: Salt Lake City International	125365	17948	7055.95	905.85	3479.08	6483.27
29	IAD	Washington, DC: Washington Dulles International	100126	17753	5460.34	629.25	5466.7	6184.61000000001
30	DAL	Dallas, TX: Dallas Love Field	77832	14995	4052.21	386.32	3808.03	6708.8
31	MDW	Chicago, IL: Chicago Midway International	81399	14569	4391.64	381.14	2771.69	6979.13
32	RDU	Raleigh/Durham, NC: Raleigh-Durham International	67762	14182	4398.6	405.53	3329.47	6030.14
33	PDX	Portland, OR: Portland International	81444	13887	4494.09	416.33	3034.22	5896.17

3. Top 10 Worst Airports

```
USE [Airline_DW]
GO
/****** Object:  StoredProcedure [dbo].[sp_Top10Worst_Report3]    Script Date: 11-04-2026
09:40:55 ******/
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO

-- Create the stored procedure
ALTER PROCEDURE [dbo].[sp_Top10Worst_Report3]
AS
BEGIN
   SET NOCOUNT ON;

   -- Select the top 10 airports with highest total delays
   SELECT TOP 10
      a.AirportCode AS Airport,
      a.AirportName,
      SUM(f.arr_flights) AS TotalFlights,
      SUM(f.arr_del15) AS TotalDelays,
      SUM(f.carrier_ct) AS CarrierDelays,
      SUM(f.weather_ct) AS WeatherDelays,
      SUM(f.nas_ct) AS NASDelays,
      SUM(f.late_aircraft_ct) AS LateAircraftDelays
   FROM dbo.Arrival_Fact f
   JOIN dbo.Dim_Airport a
      ON f.AirportKey = a.AirportKey
   GROUP BY
      a.AirportCode,
      a.AirportName
   ORDER BY
      TotalDelays DESC;
END;
```

```sql
USE [Airline_DW]
GO
/****** Object:  StoredProcedure [dbo].[sp_Top10Worst_Report3]    Script Date: 11-04-2026 09:40:55 ******/
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO

-- Create the stored procedure
ALTER PROCEDURE [dbo].[sp_Top10Worst_Report3]
AS
BEGIN
    SET NOCOUNT ON;

    -- Select the top 10 airports with highest total delays
    SELECT TOP 10
        a.AirportCode AS Airport,
        a.AirportName,
        SUM(f.arr_flights) AS TotalFlights,
        SUM(f.arr_del15) AS TotalDelays,
        SUM(f.carrier_ct) AS CarrierDelays,
        SUM(f.weather_ct) AS WeatherDelays,
        SUM(f.nas_ct) AS NASDelays,
        SUM(f.late_aircraft_ct) AS LateAircraftDelays
    FROM dbo.Arrival_Fact f
    JOIN dbo.Dim_Airport a
        ON f.AirportKey = a.AirportKey
    GROUP BY
        a.AirportCode,
        a.AirportName
    ORDER BY
        TotalDelays DESC;
END;
```

```sql
EXEC sp_Top10Worst_Report3;
```

110 % No issues found

Results | Messages

	Airport	AirportName	TotalFlights	TotalDelays	CarrierDelays	WeatherDelays	NASDelays	LateAircraftDelays
1	ORD	Chicago, IL: Chicago O'Hare International	377495	84752	24425	3398.22	31159.37	25688.11
2	DFW	Dallas/Fort Worth, TX: Dallas/Fort Worth International	344993	78407	21107.79	5677.08	22533.73	28927.53
3	DEN	Denver, CO: Denver International	344416	74734	21618.32	3039.98	29195.08	20775.67
4	ATL	Atlanta, GA: Hartsfield-Jackson Atlanta International	378642	66110	19300.51	2416.57	17703.89	26617.79
5	CLT	Charlotte, NC: Charlotte Douglas International	267713	49262	12626.29	2455.94	12421.52	21654.72
6	LAS	Las Vegas, NV: Harry Reid International	205850	40569	10872.61	1120.19	12440.47	16048.78
7	DCA	Washington, DC: Ronald Reagan Washington National	159380	39231	8376.63	1265.46	15889.92	13635.32
8	LGA	New York, NY: LaGuardia	179308	38546	9658.08	1421.73	16054.36	11352.27
9	SEA	Seattle, WA: Seattle/Tacoma International	199957	38379	10677.16	1426.86	12515.09	13557.46
10	PHX	Phoenix, AZ: Phoenix Sky Harbor International	214106	38203	13123.39	1144.48	7977.31	15880.75

4. Monthly Trends

```sql
USE [Airline_DW]
GO
/****** Object:  StoredProcedure [dbo].[sp_MonthlyTrends_Report4]    Script Date: 11-04-2026 09:43:22 ******/
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
ALTER PROCEDURE [dbo].[sp_MonthlyTrends_Report4]
AS
```

```
BEGIN
   SET NOCOUNT ON;

   SELECT
      d.Year,
      d.Month,
      SUM(f.arr_flights) AS TotalFlights,
      SUM(f.arr_del15) AS TotalDelays,
      SUM(f.carrier_ct) AS CarrierDelays,
      SUM(f.weather_ct) AS WeatherDelays,
      SUM(f.nas_ct) AS NASDelays,
      SUM(f.late_aircraft_ct) AS LateAircraftDelays
   FROM dbo.Arrival_Fact f
   JOIN dbo.Dim_Date d
      ON f.DateKey = d.DateKey
   GROUP BY
      d.Year,
      d.Month
   ORDER BY
      d.Year ASC,  -- sorts by year numerically
      d.Month ASC; -- sorts by month numerically
END;
```

```
USE [Airline_DW]
GO
/****** Object:  StoredProcedure [dbo].[sp_MonthlyTrends_Report4]    Script Date: 11-04-2026 09:43:22 ******/
SET ANSI_NULLS ON
GO
SET QUOTED_IDENTIFIER ON
GO
ALTER PROCEDURE [dbo].[sp_MonthlyTrends_Report4]
AS
BEGIN
    SET NOCOUNT ON;

    SELECT
        d.Year,
        d.Month,
        SUM(f.arr_flights) AS TotalFlights,
        SUM(f.arr_del15) AS TotalDelays,
        SUM(f.carrier_ct) AS CarrierDelays,
        SUM(f.weather_ct) AS WeatherDelays,
        SUM(f.nas_ct) AS NASDelays,
        SUM(f.late_aircraft_ct) AS LateAircraftDelays
    FROM dbo.Arrival_Fact f
    JOIN dbo.Dim_Date d
        ON f.DateKey = d.DateKey
    GROUP BY
        d.Year,
        d.Month
    ORDER BY
        d.Year ASC,  -- sorts by year numerically
        d.Month ASC; -- sorts by month numerically
END;
```

```
EXEC sp_MonthlyTrends_Report4;
```

110 % No issues found

Results | Messages

	Year	Month	TotalFlights	TotalDelays	CarrierDelays	WeatherDelays	NASDelays	LateAircraftDelays
1	2024	10	656283	84806	31555.47	1472.36	22390.29	29180.3999999999
2	2024	11	614597	89260	30151.66	2568.69	27289.08	29060.47
3	2024	12	631944	132515	43489.2900000001	5391.97	34704.27	48620.76
4	2025	1	599013	109605	35994.74	5284.38	32859.1300000001	35192.91
5	2025	2	559577	114715	34220.62	4823.42000000001	36428.1499999999	39049.7500000001
6	2025	3	664932	127648	39033.57	4596.43000000001	37271.91	46528.93
7	2025	4	644084	124749	37564.1299999999	4662.5	36994.32	45297.06
8	2025	5	667586	155262	43212.07	7216.09	49775.12	54855.0200000001
9	2025	6	674179	186766	55228.28	8968.33999999999	50276.82	71961.68
10	2025	7	696049	192888	57600.7200000001	9662.76000000001	50585.0500000001	74730.62
11	2025	8	666242	145517	45479.29	6893.64000000001	38190.52	54683.94
12	2025	9	621601	100369	32170.54	3670.33	30019.85	34303.28
13	2025	10	668332	132452	40207.62	2699.07	41161.0300000001	48171.99

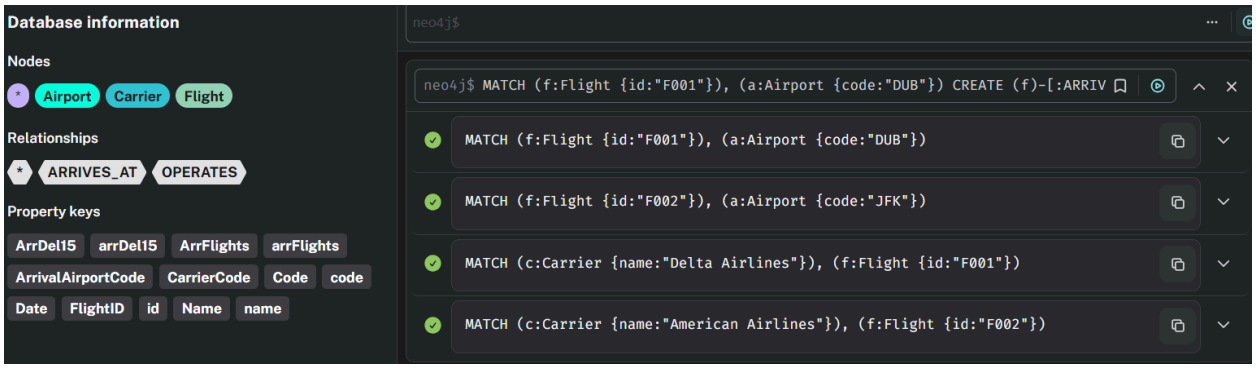
APPENDIX B - NEO4J CODE

1. Create Nodes

```
CREATE (a1:Airport {code:"DUB", name:"Dublin Airport"})
CREATE (a2:Airport {code:"JFK", name:"John F. Kennedy Airport"})

CREATE (c1:Carrier {name:"Delta Airlines"})
CREATE (c2:Carrier {name:"American Airlines"})

CREATE (f1:Flight {id:"F001", arrFlights:120, arrDel15:30})
CREATE (f2:Flight {id:"F002", arrFlights:200, arrDel15:50});
```

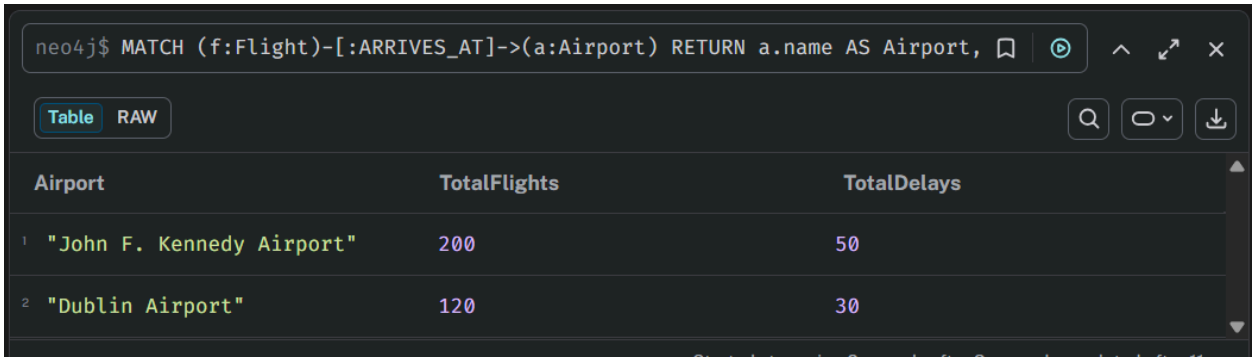
2. Create Relationships

```
MATCH (f:Flight {id:"F001"}), (a:Airport {code:"DUB"})
CREATE (f)-[:ARRIVES_AT]->(a);

MATCH (f:Flight {id:"F002"}), (a:Airport {code:"JFK"})
CREATE (f)-[:ARRIVES_AT]->(a);

MATCH (c:Carrier {name:"Delta Airlines"}), (f:Flight {id:"F001"})
CREATE (c)-[:OPERATES]->(f);

MATCH (c:Carrier {name:"American Airlines"}), (f:Flight {id:"F002"})
CREATE (c)-[:OPERATES]->(f);
```



3. Analysis Query - Airport Performance

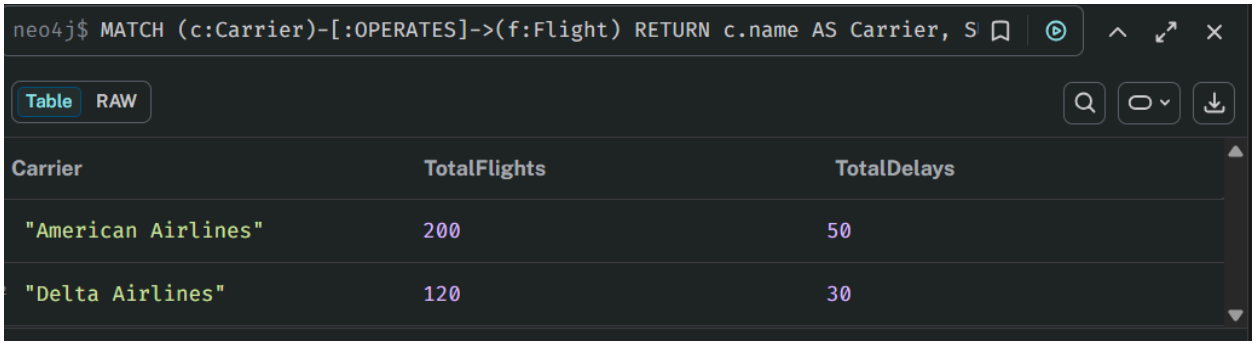
```
MATCH (f:Flight)-[:ARRIVES_AT]->(a:Airport)
RETURN a.name AS Airport,
    SUM(f.arrFlights) AS TotalFlights,
    SUM(f.arrDel15) AS TotalDelays
ORDER BY TotalDelays DESC;
```



	Airport	TotalFlights	TotalDelays
1	"John F. Kennedy Airport"	200	50
2	"Dublin Airport"	120	30

4. Analysis Query - Carrier Performance

```
MATCH (c:Carrier)-[:OPERATES]->(f:Flight)
RETURN c.name AS Carrier,
    SUM(f.arrFlights) AS TotalFlights,
    SUM(f.arrDel15) AS TotalDelays
ORDER BY TotalDelays DESC;
```

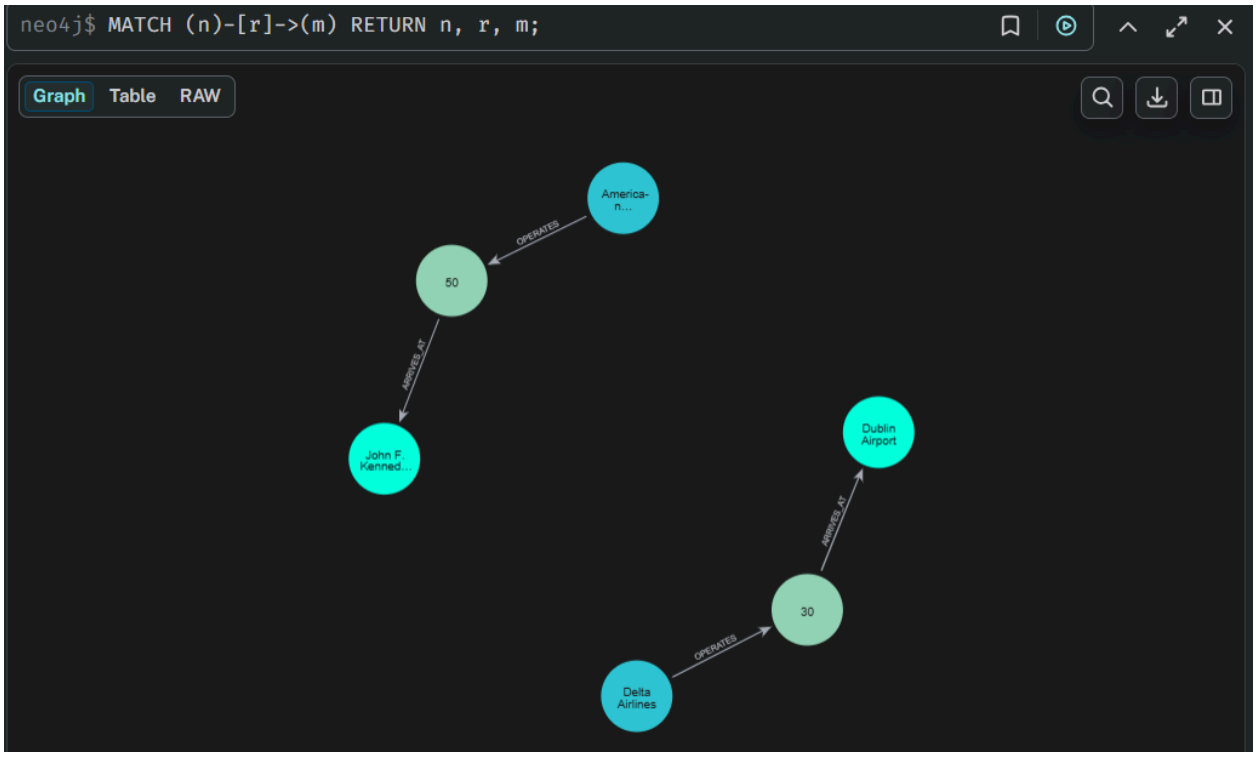


Carrier	TotalFlights	TotalDelays
"American Airlines"	200	50
"Delta Airlines"	120	30

5. Relationship Validation Query

```
MATCH (n)-[r]->(m)
```

RETURN n, r, m;


neo4j$ MATCH (n)-[r]->(m) RETURN n, r, m;
Graph
Table
RAW
America-n...
OPERATES
50
ARRIVES_AT
John F. Kenned...
Dublin Airport
ARRIVES_AT
30
OPERATES
Delta Airlines